

REGEX RULE ENGINE

How fee-schedule comments become review-ready rule changes

The engine extracts known facts, compares them with existing rules, and produces a controlled decision for human review.

01

Pattern library

About 25 regex patterns target rates, dates, age limits, authorization, specialty codes, service locations, frequency limits, methodology terms, change types, citations, and rental periods.

02

Fact extraction

Each match records the field name, value, raw matched text, pattern name, and a fixed high, medium, or low confidence level.

03

Rule comparison

The engine checks CSRA condition tables, free-text SOP rules, or row-level data when the first two structures do not match.

04

Review output

Batch processing returns a summary DataFrame and draft SQL INSERT or UPDATE statements. The engine never executes SQL automatically.

ENGINE PIPELINE

1

Read comment

Receive the fee-schedule analyst comment.

2

Extract facts

Identify rates, dates, limits, requirements, and change language.

3

Compare rules

Check extracted facts against the available rule structure.

4

Prepare review

Return a decision, confidence, suggested changes, and draft SQL.

DECISION OUTCOMES

NO CHANGE

No actionable difference, or the comment explicitly says unchanged.

NEW RULE

No existing rule exists, so each extracted fact becomes a proposed entry.

UPDATE

A fact differs from the current value and the comment contains clear change language.

CONFLICT

Values differ, but the change language remains inconclusive and requires review.

Fee-schedule rule automation for consistent business review

BUSINESS PURPOSE

- Standardize comment review**
Convert recurring analyst language into structured facts that the business can review consistently.
- Identify the required action**
Classify each comment as no change, a new rule, an update, or a conflict that needs attention.
- Prepare the change package**
Present the current value, proposed value, reason, evidence, confidence, and draft SQL together.

WHAT THE BUSINESS RECEIVES

- A clear decision and confidence level for every comment
- Suggested changes with field, old value, new value, and reason
- A readable summary linked to the matched source text
- Batch results in a summary DataFrame
- Draft INSERT or UPDATE statements for reviewer approval

CONTROL FRAMEWORK

Control	How the solution handles it	Business safeguard
Fact capture	Extracts rates, dates, limits, authorization requirements, codes, and change language	Uses defined fields instead of free-form interpretation
Evidence	Stores the raw matched text and the pattern that produced each fact	Keeps every result traceable to the source comment
Rule check	Compares facts with CSRA conditions, SOP rule text, or available row data	Checks the proposed value against existing business rules
Decision	Returns no change, new rule, update, or conflict	Routes unclear cases for focused review
Confidence	Aggregates the confidence assigned to extracted facts	Helps reviewers prioritize validation
Execution	Generates draft SQL as text only	Nothing runs automatically in the database

Operating model: The solution prepares a complete rule-change proposal. A reviewer validates the evidence and approves the change before any downstream execution.